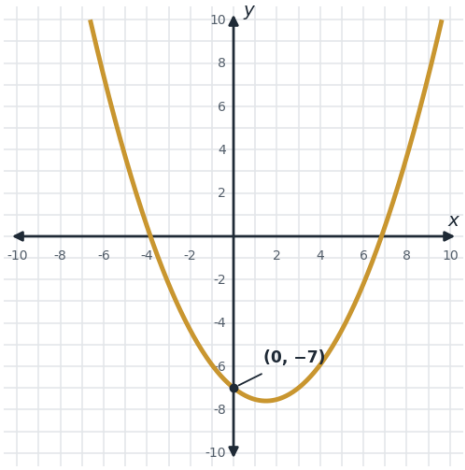
Question 1

Consider the following formula:

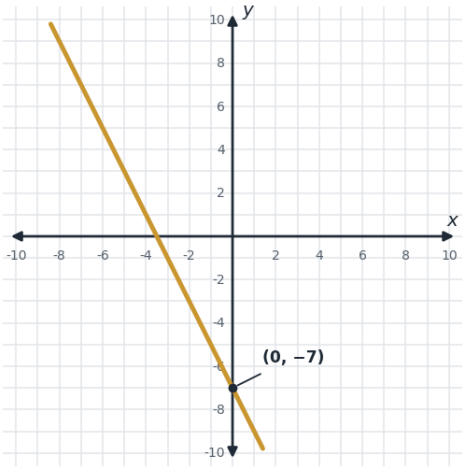
$$6y = 12x - 42$$

Which of the following choices represents the same relationship as the given function?

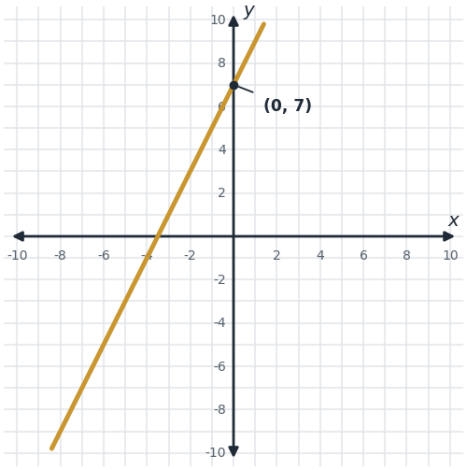
A)



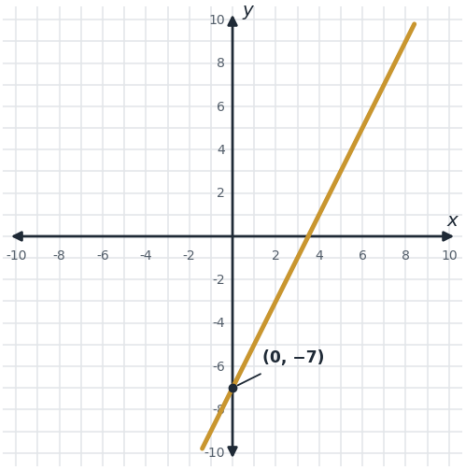
B)



C)



D)



WORKSPACE

Question 2

What is the next number in the following series of numbers?

$-3.5, \quad -2.75, \quad -2, \quad -1\frac{1}{4}, \quad ?$

A) -0.75

B) -0.5

C) 0.25

D) 0.5

WORKSPACE

Question 3

Simplify the following expression:

$$h = (3R^2 + 4u - t) + \frac{-3R^3 + 2Ru + 5Rt}{R}$$

A) $h = 6R^2 + 6u + 4t$

B) $h = 6u + 4t$

C) $h = 6u - 6t$

D) $h = -6R^2 + 6u + 4t$

WORKSPACE

Question 4

Evaluate the expression when:

$y = 3, \quad w = -2, \quad h = \frac{1}{2}$

$$\frac{2y - 3w + h}{4}$$

A) $\frac{5}{2}$

B) $\frac{11}{4}$

C) $\frac{25}{8}$

D) $\frac{7}{2}$

WORKSPACE

Question 5

Factor the expression:

$$18ge^7 - 24ge^2n^4$$

A) $6ge^2(3e^5 - 4n^4)$

B) $6e(3e^6 - 4en^4)$

C) $3ge^2(6e^5 - 8n^4)$

D) $6ge^2(3e^5 + 4n^4)$

WORKSPACE

Question 6

Solve for x:

$$3x^2 - 7x - 11 = 0$$

A) $x = \frac{-7 \pm \sqrt{181}}{6}$

B) $x = \frac{7 \pm \sqrt{181}}{3}$

C) $x = \frac{-7 \pm \sqrt{49}}{6}$

D) $x = \frac{7 \pm \sqrt{181}}{6}$

WORKSPACE

Question 7

What is the next number in the following series of numbers?

4, 27, 8, 22, 16, 17, 32, ?

A) 10

B) 12

C) 15

D) 64

WORKSPACE

Question 8

Find the next number in the series:

$\frac{1}{2}, \frac{9}{8}, \frac{7}{4}, \frac{19}{8}, ?$

A) $\frac{5}{2}$

B) $\frac{21}{8}$

C) 3

D) $\frac{25}{8}$

WORKSPACE

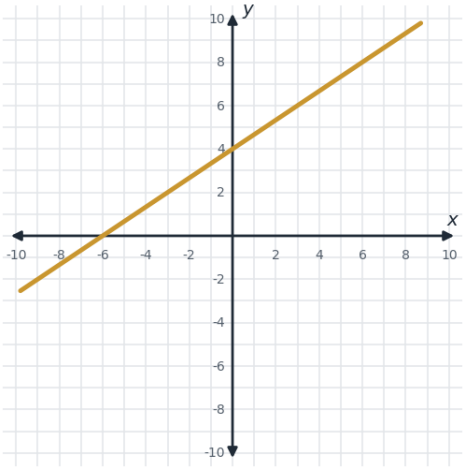
Question 9

Consider the following formula:

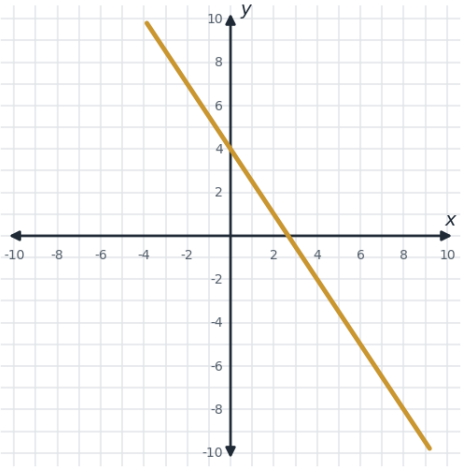
$$y = -\frac{2}{3}x + 4$$

Which of the following choices represents the same relationship as the given function?

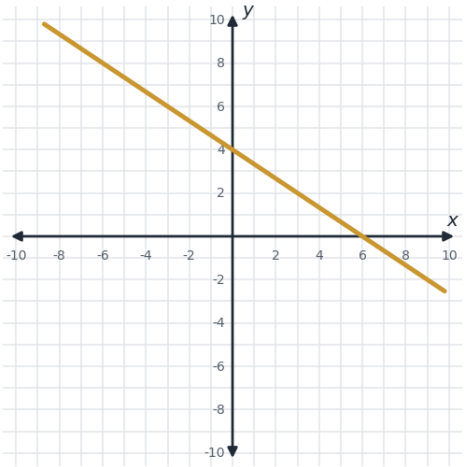
A)



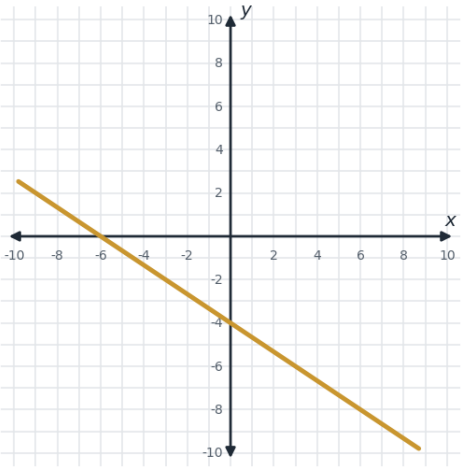
B)



C)



D)



WORKSPACE

Question 10

Consider the following formula:

$$y = -\frac{1}{2}x + 4$$

Which of the following choices represents the same relationship as the given function?

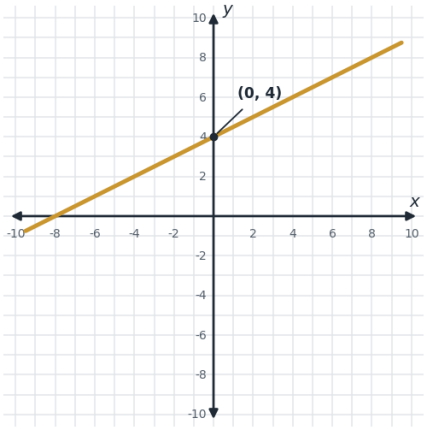
A)

x	y
−6	8
−4	7
−2	6
2	4
4	3

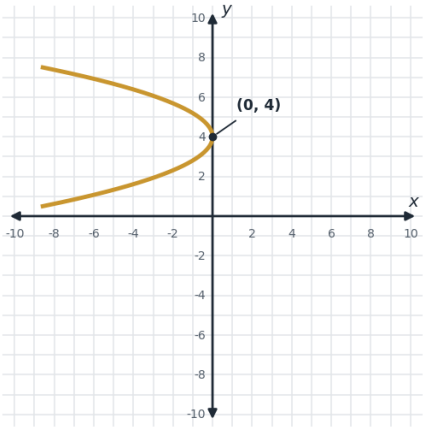
B)

x	y
−8	8
−4	6
2	3
6	1
10	−1

C)



D)



WORKSPACE

Question 11

Use the table below to determine the linear relationship:

x	y
0	9
1	7
2	5
3	3
4	1

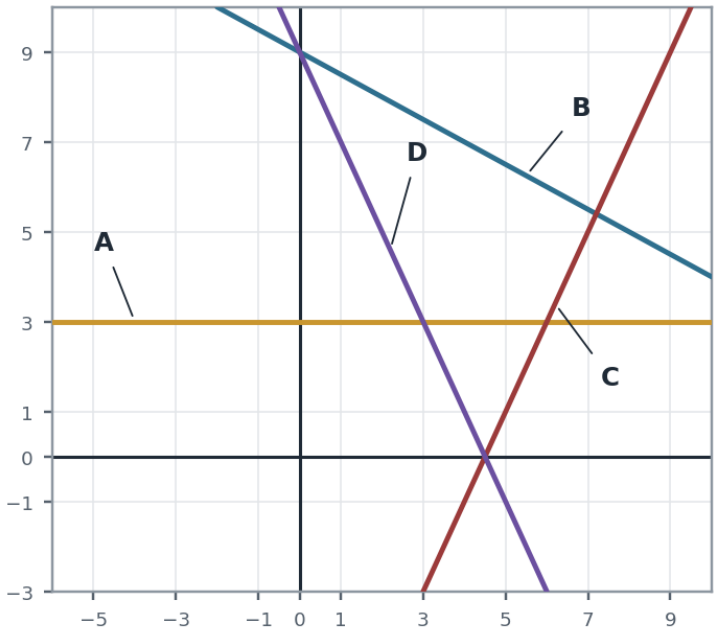


Figure 1

Which graph represents the same relationship in Figure 1?

A) Line A

B) Line B

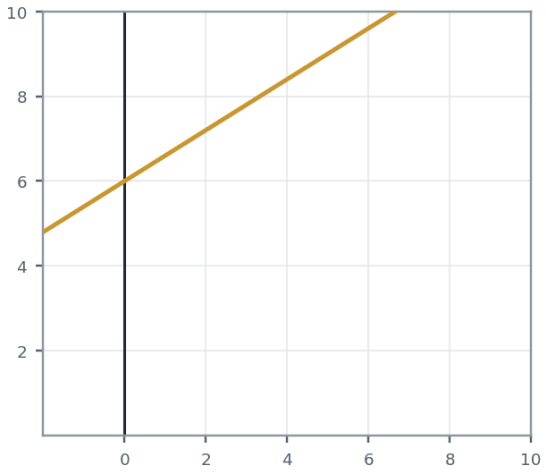
C) Line C

D) Line D

WORKSPACE

Question 12

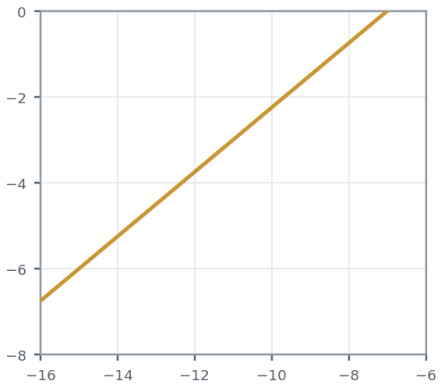
Consider the following graph of the function $y = mx + b$:



Which of the following options is true regarding this graph?

A) Function A, $y = -\frac{3}{5}x + 6$, has the same slope and the same y-intercept.

B) Function B, $y = \frac{3}{5}x + 6$, has the same slope and the same y-intercept.

C) 

D) Function D, $y = \frac{3}{5}x - 6$, has the same slope and the same x-intercept.

Question 13

Consider the table to answer the question below:

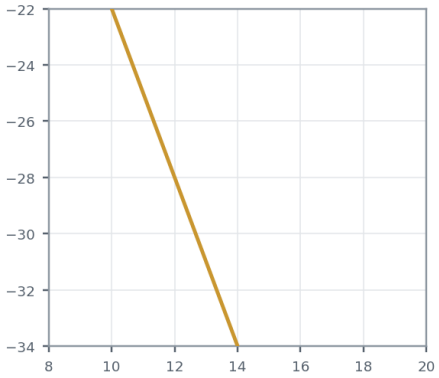
x	y
14	50
27	89
42	134
49	155
63	197

Which of the following choices represents the same relationship as the given table?

A)

x	y
−9	−28
−6	−16
−4	−8
−1	4
3	20

B)



C)

x	y
11	41
24	80
43	137
47	149
66	206

D) Function D has a y-intercept of 8 and increases by 1 for every 3 units that x increases.

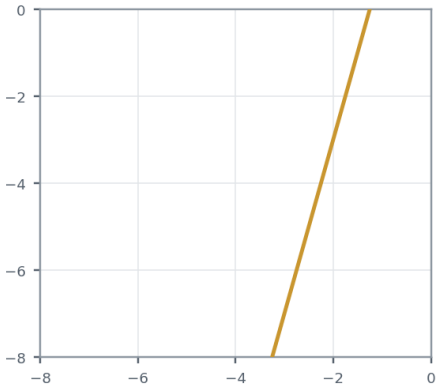
WORKSPACE

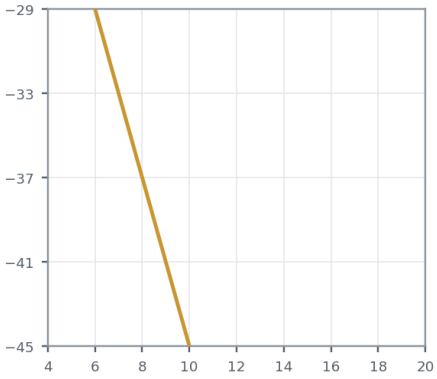
Question 14

Translate the statement below, then answer the question.

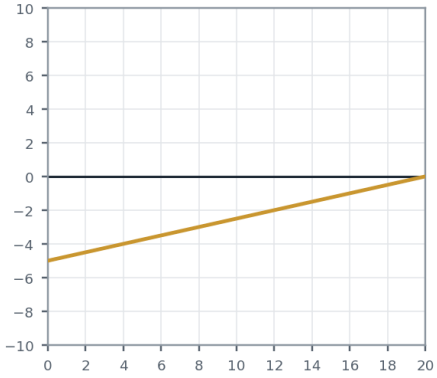
The value of y is 5 less than 4 times the value of x.

Which of the following choices represents the same relationship as the equation above?

A) 

B) 

C) None of the answer choices are correct.

D) 

WORKSPACE

Question 15

Given the equation:

$$m = -3k + 8$$

Which statement is true for this formula?

A) If k increases, then m increases.

B) If k increases, then m decreases.

C) If k increases, then m stays constant.

D) If m increases, then k stays constant.

WORKSPACE

Question 16

Given the equation:

$y = -2x + 7.$

Which statement is true for this formula?

A) If x increases from 1 to 4, then y increases.

B) If x decreases from 4 to 1, then y decreases.

C) If y increases from -1 to 5, then x increases.

D) If y increases from -1 to 5, then x decreases.

WORKSPACE

Question 17

Consider the following formula:

$$x = 4(y - 6) - 3$$

Which statement is true for this formula?

A) When the value of y is greater than 6, x is positive.

B) When the value of y is less than 7, x is positive.

C) When the value of y is greater than 7, x is positive.

D) When the value of y is less than 6, x is positive.

WORKSPACE

Question 18

Consider the following formula:

$r = 18 - 4t$

Which of the following statements is true for this formula?

A) If t increases, then r increases.

B) If t increases, then r decreases.

C) If r increases, then t increases.

D) If t increases, then r stays the same.

WORKSPACE

Question 19

Consider the following formula:

$$p = 16 - 2q$$

Which of the following statements is true for this formula?

A) If p is less than 0, then q is greater than 8.

B) If p is greater than 0, then q is greater than 8.

C) If q is less than 8, then p is negative.

D) If q is greater than 8, then p is positive.

WORKSPACE

Question 20

Expand the expression:

$$-3(3x-5)(2x+7)$$

A) $-18x^2-33x+105$

B) $-18x^2-63x+105$

C) $-18x^2+30x-105$

D) $18x^2+33x-105$

WORKSPACE

Question 21

Factor the expression:

$x^2 + 7x + 12$

A) $(x + 2)(x + 6)$

B) $(x + 1)(x + 12)$

C) $(x + 3)(x + 4)$

D) $(x - 3)(x - 4)$

WORKSPACE

Question 22

Factor the expression:

$6x^2 + 11x + 3$

A) $(6x + 1)(x + 3)$

B) $(3x + 3)(2x + 1)$

C) $(6x - 1)(x - 3)$

D) $(3x + 1)(2x + 3)$

WORKSPACE

Question 23

Given the equation:

$$a = -\frac{7}{9}ct \qquad c = \frac{3s - 6}{2}$$

If $a = -56$ and $s = 8$, what is the value of t?

A) 6

B) 8

C) 9

D) 12

WORKSPACE

Question 24

Given the equation:

$$r = -\frac{3}{5}(2k - 4) + \frac{1}{2}a - m$$

If $r = 12$, $a = -6$, and $m = -3$, what is the value of k?

A) -6

B) -8

C) 8

D) 12

WORKSPACE

Question 25

Simplify the following expression:

$$3(x-2)^2 - 2(x+4)^2$$

A) $x^2 - 28x - 20$

B) $x^2 - 28x + 20$

C) $x^2 + 28x - 20$

D) $5x^2 - 28x - 20$

WORKSPACE

Question 26

Evaluate the formula:

$$Q = \frac{5}{r}$$

$$r = \frac{2n - n}{10\pi}$$

A) $\frac{10\pi}{5n}$

B) $\frac{5n}{10\pi}$

C) $\frac{50\pi}{n}$

D) $\frac{n}{50\pi}$

WORKSPACE

Question 27

Isolate the variable from the formula:

$$J_1 = \frac{C_1(J_2 + C_2) - C_1 J_2}{C_2 - 3}$$

$C_1 = ?$

A) $C_1 = \frac{J_1(C_2 + 3)}{C_2}$

B) $C_1 = \frac{J_1(C_2 - 3)}{C_2}$

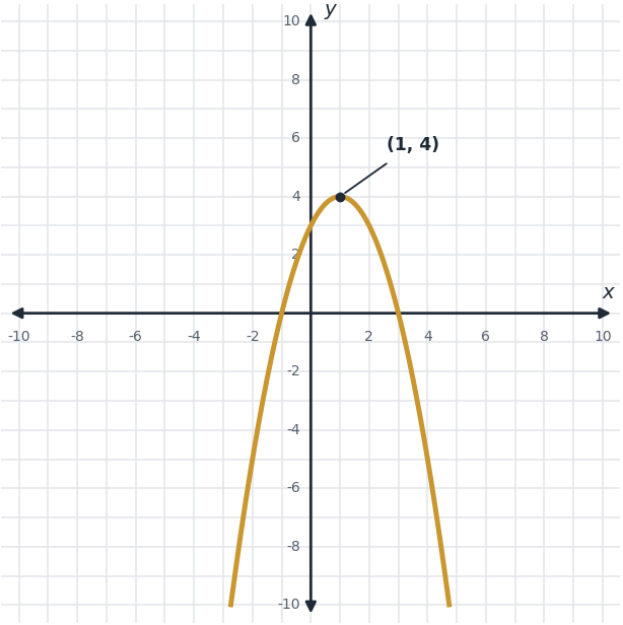
C) $C_1 = \frac{C_2 - 3}{J_1 C_2}$

D) $C_1 = \frac{J_1 C_2}{C_2 - 3}$

WORKSPACE

Question 28

Which equation matches the graph shown?



A) $y = -(x + 1)^2 + 4$

B) $y = -(x - 1)^2 - 4$

C) $y = -(x + 1)^2 - 4$

D) $y = -(x - 1)^2 + 4$

WORKSPACE

Question 29

Consider the following formula:

$$s = -2(3t - 4) + 16$$

Which of the following statements is NOT true for this formula?

A) If $t > 4$, then s is negative.

B) If $t < 4$, then s is positive.

C) If $s = 0$, then $t = 4$.

D) If s is negative, then $t < 4$.

WORKSPACE

Question 30

Consider the following formula:

$$n = 30 - 5r$$

Which of the following statements are true?

I. If r increases by 2, then n decreases by 10.

II. If n is less than 0, then r is greater than 6.

III. If $r = 5$, then $n = 0$.

A) I only

B) II and III only

C) I and II only

D) All statements are correct

WORKSPACE

Question 31

Consider the following formula:

$$P = 3Q + 6$$

Which of the following statements is true for this formula?

A) The difference between the values of P and Q changes.

B) The difference between the values of P and Q is constant.

C) For positive P values, $Q > 0$.

D) For negative Q values, $P < 0$.

WORKSPACE

Question 32

Solve for x:

$$\frac{x-3}{6}=\frac{5}{8}$$

A) 6.75

B) 7.25

C) 6

D) 9

WORKSPACE

Question 33

Solve the systems of equations:

$$2x + 3y = 17$$

$$6x - 5y = -5$$

A) $(4,\ \frac{5}{2})$

B) $(\frac{5}{2},\ 4)$

C) $(\frac{5}{2},\ -4)$

D) $(3,\ 4)$

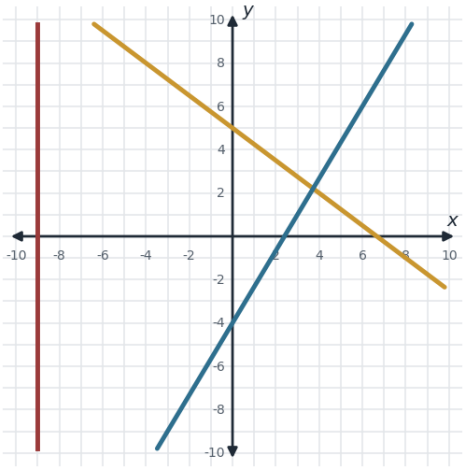
WORKSPACE

Question 34

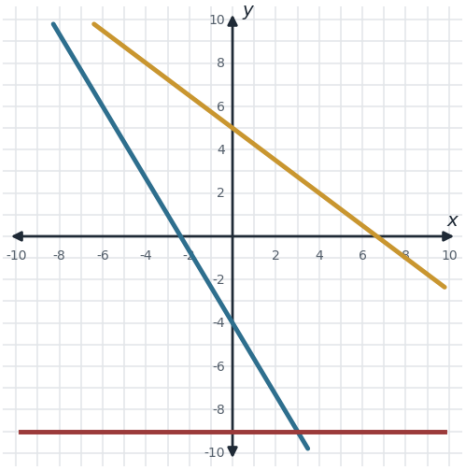
Which graph represents the following equations?

$y = -\frac{3}{4}x + 5$ $y = \frac{5}{3}x - 4$ $y = -9$

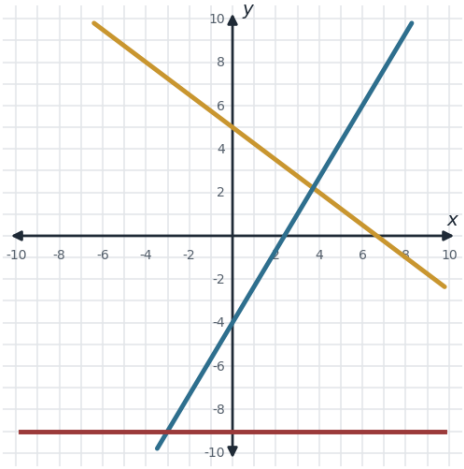
A)



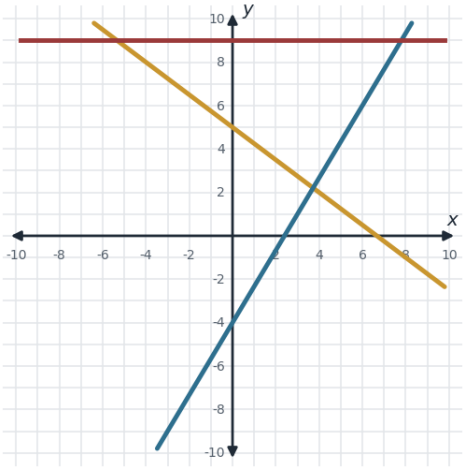
B)



C)



D)



WORKSPACE

Answer Key

1. D Divide by 6: y = 2x − 7. Line crossing the y-axis at −7 and rising 2 per step.

2. B Add 0.75 each time: −1.25 + 0.75 = −0.5.

3. B Divide each top term by R: $-3R^2 + 2u + 5t$. Add to $3R^2 + 4u - t \rightarrow 6u + 4t$.

4. C (6 + 6 + ½) / 4 = 12.5 / 4 = 25/8.

5. A Greatest common factor is $6ge^2$. C multiplies back correctly but only pulls out $3ge^2$, so it isn't fully factored.

6. D Quadratic formula: (7 ± √(49 + 132)) / 6 = (7 ± √181)/6.

7. B Two patterns: 4, 8, 16, 32 (×2) and 27, 22, 17 (−5). Next is 17 − 5 = 12.

8. C In eighths: 4, 9, 14, 19 (+5). Next 24/8 = 3.

9. C Slope −2/3 (falls), crosses y-axis at 4 and x-axis at 6.

10. B Check rows: −½(−8)+4 = 8, −½(6)+4 = 1 … all match B. A is y = −½x + 5.

11. D Table: y = 9 − 2x. Line D starts at 9 and hits 0 at x = 4.5.

12. B Graph: slope 3/5, y-intercept 6. Function B is the identical line.

13. C Table: y = 3x + 8. Every row of C fits (3·11 + 8 = 41 …). A is y = 4x + 8.

14. C y = 4x − 5. A is y = 4x + 5, B is y = −4x − 5, D is y = ¼x − 5 — none match.

15. B Negative coefficient on k: as k goes up, m goes down.

16. D y = −2x + 7: y going up means x going down.

17. C x = 4y − 27 = 0 at y = 6.75. Above 7, x is always positive.

Answer Key (cont.)

18. B Negative coefficient on t: t up → r down.

19. A p = 0 at q = 8. p negative ⇔ q greater than 8.

20. A $(3x - 5)(2x + 7) = 6x^2 + 11x - 35$; × (−3) = $-18x^2 - 33x + 105$.

21. C 3 × 4 = 12 and 3 + 4 = 7.

22. D $(3x + 1)(2x + 3) = 6x^2 + 9x + 2x + 3$.

23. B c = (24 − 6)/2 = 9. Then −56 = −(7/9)(9)t = −7t → t = 8.

24. B ½(−6) − (−3) = 0, so 12 = −(3/5)(2k − 4) → 2k − 4 = −20 → k = −8.

25. A $3x^2 - 12x + 12 - 2x^2 - 16x - 32 = x^2 - 28x - 20$.

26. C r = n / (10π). Q = 5 ÷ r = 50π / n.

27. B Top simplifies to C_1C_2, so $J_1 = C_1C_2/(C_2 - 3) \rightarrow C_1 = J_1(C_2 - 3)/C_2$.

28. D Vertex (1, 4), opens down: $y = -(x - 1)^2 + 4$.

29. D s = −6t + 24 → zero at t = 4. s negative means t > 4, so D is false.

30. C I and II are true. r = 5 gives n = 5, not 0.

31. A P − Q = 2Q + 6, which changes as Q changes. C fails at Q = −1 (P = 3).

32. A x − 3 = 30/8 = 3.75 → x = 6.75.

33. B Triple the first: 6x + 9y = 51. Subtract: 14y = 56 → y = 4, x = 5/2.

34. C Only C has a falling orange line through 5, a rising blue line through −4, and a horizontal red line at y = −9.